

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 201 – INTRODUCTION TO THERMODYNAMICS
Fall 2019**

Lecture:

MWF: 11:00am – 11:50am
Room: EIB 124

Office Hours:

12pm-1pm MW by appointment (Dr. Bilgin)
..... T/Th by appointment (TAs)

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teach Assistants:

Discussion periods: Tuesday/Thursday: 9:30 – 10:20 AM

Text:

***Fundamentals of Engineering Thermodynamics*, 9th ed., Moran, Shapiro, Boettner and Bailey (Hardcover, Looseleaf, Ebook can be used)**

NOTE: MUST USE 9th Edition of textbook

Course Description:

1st Law of thermodynamics; work, heat and conversion of energy in closed and open systems, including engineering components, power and refrigeration cycles. Engineering problem-solving methodology, pure substance properties, property tables, equations of state, theory of gases and other states of matter. Second Law of thermodynamics; entropy, open and closed system entropy balances and equilibrium, with applications. Fundamental thermodynamic functions, Maxwell relations, Clapeyron equation.

Course Aims, Scope, Level:

The engineering science courses that form the basis of chemical engineering can be roughly divided into two sequences: transport processes and thermodynamics. This course is the first in the thermodynamics sequence and is designed to give the student a firm grounding in energy and entropy concepts. Later courses will deal in more detail with phase and reaction equilibrium. Knowledge of thermodynamics is needed in subsequent undergraduate chemical engineering courses. If a student can master the fundamentals early in the course of study, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have the necessary background in elementary mathematics, and this mathematical background will be used. Also, students should understand mechanics which is usually taught in the first college physics course.

By the end of the term, the students will understand energy and entropy concepts for closed systems and open systems (control volumes). Student will be able to apply engineering methodology to a variety of situations in order to solve thermodynamic problems. Energy and entropy are conceptually difficult and require time and effort on the student's part before they are well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as possible, and thoughtful questions should be brought up during lecture and discussion. Good luck!

Course Outline:

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 4)
- VI. The Second Law of Thermodynamics (Chapter 5)
- VII. Entropy (Chapter 6)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

Exams, Homework and Grading

There will be 4 to 6 quizzes, two midterm exams, and one final exam. All exams are required. There will be three 50 minute-lectures per week (Mon-Wed-Fri) and one, TA-led discussion session on Tuesday or Thursday. New material will be covered in lecture. Students should stay current with the material and be prepared to answer questions. During the discussion sessions, homework problems will be worked and occasionally new material covered. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care.

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for CHE 201:

- The importance of Process Safety
- Hazard Recognition

Grading Policy:

Course grades will be weighted according to the following grading formula.

Two exams (20% each)	40%
Final exam	20%
Quizzes	5%

Homework problems	15%
Attendance	5%
Project	15%

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

Additional Information and Policies

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies.
2. Cheating on homework, quizzes, and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Quizzes will be held periodically during the class period. There will be no makeup quizzes.
4. Late Homework Policy: Homework can be turned in 1 lecture period late without excuse for up to 50% credit.
5. Homework format guidelines are given in Blackboard/Course Content.
6. Homework/quiz grading: Lowest homework and quiz grade will be dropped.
7. Lecture Mobile Technology Policy will be given in 1st Lecture period.